Reproductive Health

Complete Study Notes: Problems, Strategies, and Medical Interventions.

Reproductive Health: Programmes & Aims

1951 → Family Planning Initiated → **Reproductive & Child Health Care (RCH)**

- Create awareness among people about various reproduction-related aspects.
- Provide facilities and support for building up a **reproductively** healthy society.

Action Strategies for Awareness

- **Media Support:** Audio-visuals & print media utilized by governmental and non-governmental agencies.
- **Sex Education in Schools:** Introduced to provide right info to young, actively discouraging children from believing in myths and misconceptions.
- **Adolescent Education:** Providing proper info about reproductive organs, adolescence-related changes, safe and hygienic sexual practices, STDs, and AIDS.
- **Fertile Couples & Marriageable Age Groups:** Educated on birth control options, care of pregnant mothers, post-natal care of mother and child, importance of breastfeeding, and equal opportunities for male and female children.

Key Goal:
Building a socially responsible & healthy society!

Medical Assistance & Amniocentesis

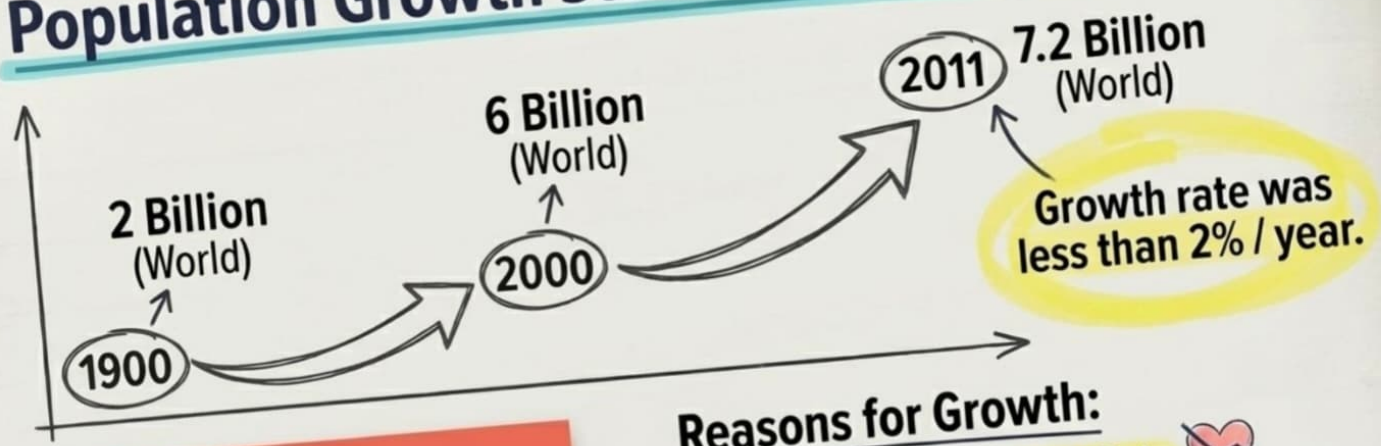
Medical Assistance Provided For:

- Pregnancy
- Delivery
- STDs
- Abortions
- Contraception
- Menstrual problems
- Infertility

WARNING: Amniocentesis

- **Definition:** Analyzing fetal cells and dissolved substances from amniotic fluid.
- **Statutory Ban:** Strictly banned for sex determination.
- **Valid Medical Uses:** Testing for genetic disorders such as Down syndrome, haemophilia, and sickle-cell anaemia.

Population Growth Statistics & Trends



Alarming growth could lead to an absolute scarcity of basic requirements: Food, Shelter, & Clothing.

Reasons for Growth:

- Rapid decline in Death Rate
- Decline in Maternal Mortality Rate (MMR)
- Decline in Infant Mortality Rate (IMR)
- Increase in number of people in reproducible age.

Overcoming Population Growth

Motivation via Contraceptives: Encourage smaller families.

Hum Do Hamare Do

(Note: Many couples have even adopted a 'one child norm').

Statutory Age Raised

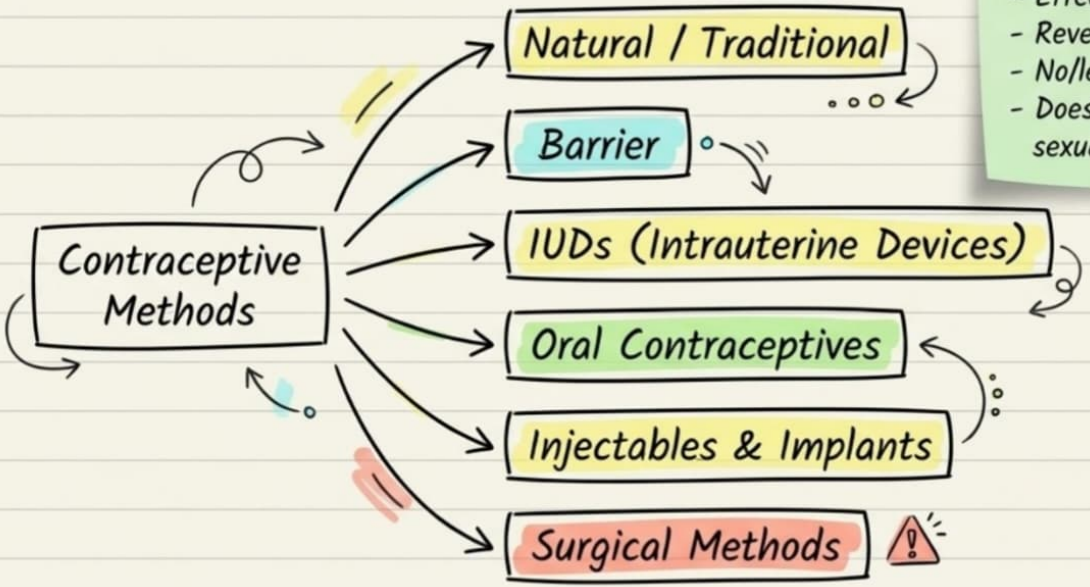
18 for females, 21 for males

Incentives

Financial or social incentives given to couples with small families.

Introduction to Contraceptives

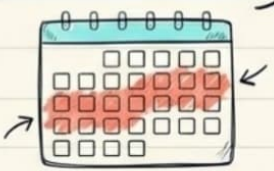
Ideal Contraceptive Traits:
- User-friendly
- Easily available
- Effective
- Reversible
- No/least side effects
- Doesn't interfere with sexual drive/desire



Natural / Traditional Methods

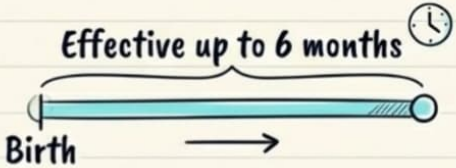
Principle: Works on the principle of avoiding chances of ovum and sperm meeting.

1. Periodic Abstinence: Abstain from coitus from Day 10 to 17 of the menstrual cycle.



2. Withdrawal (Coitus Interruptus): Withdraw penis from vagina before ejaculation to avoid insemination.

3. Lactational Amenorrhea: Ovulation and the menstrual cycle do not occur during intense lactation following parturition.



Barrier Methods

Prevents the entry of semen into the vaginal tract. Made up of thin rubber/latex sheaths.

Condoms (Nirodh)

Diaphragms

Cervical Caps

Vaults

Key Enhancements:

- Major Bonus: Prevents infection of STIs and AIDS.
- Efficiency Boost: Spermicidal creams, jellies, and foams are usually used alongside these barriers to increase contraceptive efficiency.

Intrauterine Devices (IUDs)

Non-Medicated IUDs	Copper-Releasing IUDs	Hormone-Releasing IUDs
Example: Lippes loop.	Examples: CuT, Cu7, Multiload 375.	Examples: Progestasert, LNG-20.
	Mechanism: They release Cu^{++} ions which suppress sperm motility and their fertilizing capacity.	**Use Case:** Ideal contraceptives for females who want to delay pregnancy and/or space children.

Oral Pills, Injectables & Emergency Contraceptives

Standard Oral Pills:
Small doses of progestogens (or combos). Inhibit ovulation/implantation and alter cervical mucus.

Take for 21 days → 7-day gap (menses) → Repeat.

'Saheli' Pill:
Non-steroidal preparation, once-a-week pill.

Injectables & Implants:
Progestogens used by females as injections or implants under the skin.

Emergency Contraception:
Oral contraceptives must be taken within 72 hours of coitus. Used for casual unprotected intercourse, rapes, or contraceptive failure.

Surgical Methods (Sterilisation)

Core Principle: Block gamete transport and permanently prevent conception. Highly effective.

Vasectomy (Males)

- ✓ A small part of the vas deferens is removed or tied up.
- ✓ Procedure done through a small incision on the scrotum.

Tubectomy (Females)

- ✓ A small part of the fallopian tube is removed or tied up.
- ✓ Procedure done through a small incision in the abdomen or through the vagina.

Medical Termination of Pregnancy (MTP)

Definition: Intentional or voluntary termination of pregnancy before full term (induced abortion). Legalised by Govt of India in 1971 with strict rules to avoid misuse.

Why is it done?: To get rid of unwanted pregnancies (unprotected coitus, contraceptive failure, rapes) or in cases of harmful/fatal pregnancies.

Warning: MTPs done illegally by unqualified doctors are unsafe and can be fatal. Strict laws exist to prevent sex determination leading to female foeticide.

Sexually Transmitted Infections (STIs)

Definitions:

Diseases transmitted through sexual intercourse. Also known as Venereal Diseases (VD) or Reproductive Tract Infections (RTI).

Master List of Infections:

Gonorrhoea, Syphilis, Chlamydiasis, Genital Warts, Trichomoniasis

The Incurables

- Hepatitis-B
- HIV
- Genital Herpes

Transmission Paths for Hepatitis-B & HIV:

- Sharing injection needles
- Blood transfusions
- Sharing surgical instruments
- Infected mother to foetus

STI Progression & Prevention

Highly prone to age groups 15-24 yrs!

Early Symptoms: Minor itching, fluid discharge, slight pain, swelling. (Note: Infected females **are often** asymptomatic and remain undetected).

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Deterrence: Mild symptoms deter timely detection and proper treatment.

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Severe Complications: Leads to Pelvic Inflammatory Diseases (PID), abortions, stillbirths, ectopic pregnancies, infertility, or even cancer of the reproductive tract.

Prevention (The Rule of 3):

1. Avoid sex with unknown/multiple partners.
2. Use condoms during coitus.
3. Consult a doctor for early detection in case of doubt.

Infertility & Introduction to IVF

Infertility Overview: Many couples are unable to conceive due to health issues. Specialized health care units (infertility clinics) help in diagnostics and corrective treatment.

The ART Pivot: When corrections are not possible, couples are assisted through Assisted Reproductive Technologies (ART).

In Vitro Fertilization (IVF)

- **Definition:** Fertilization outside the body in conditions similar to natural fertilization. Popularly known as the 'test tube baby program'.
- **Step 1:** Ova from female and sperm from male are collected and induced to form a zygote.

(**Note:** Embryos formed by in-vivo fertilization—fusion of gametes within the female-can also be used to assist females who cannot conceive).

Embryo Transfer Protocols

Zygote / Early Embryo

Up to 8 Blastomeres

ZIFT

(Zygote Intra Fallopian Transfer)

Transferred to Fallopian Tube

More than 8 Blastomeres

IUT

(Intra Uterine Transfer)

Implanted directly into Uterus

Advanced ART Techniques

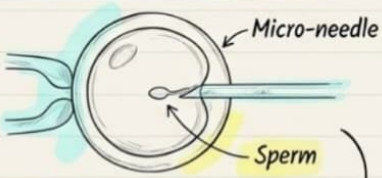
GIFT (Gamete Intra Fallopian Transfer)

Transfer of ovum collected from a donor into the fallopian tube of another female.

↳ Use Case: For females who cannot produce an ovum but can provide a suitable environment for fertilization and development.

ICSI (Intra Cytoplasmic Sperm Injection)

A specialized method where a sperm is directly injected into the ovum.



AI (Artificial Insemination)

Semen collected from the husband or a healthy donor is artificially introduced into the vagina or uterus (IUI – Intra Uterine Insemination).

↳ Use Case: Helpful if the male partner is unable to inseminate or has a low sperm count.

☆ End of Notes - Review thoroughly! ✓